HSR-PE-07 BGP Route Churn and OSPF Adjacency Loss Troubleshooting

Node: HSR-PE-07 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_1647

Description

BGP route churn and OSPF adjacency loss on the PE router HSR-PE-07 at the HSR Layout Access site, running Cisco IOS, is a critical failure domain that originates from the inherent instability of BGP route reflection and the potential for OSPF LSA flooding. The Cisco PE router operates in a multi-protocol environment where BGP (Border Gateway Protocol) and OSPF (Open Shortest Path First) protocols coexist. BGP route reflection is a mechanism that enables a BGP route reflector to advertise routes learned from its client neighbors to its other client neighbors, which can lead to a rapid propagation of route changes in the event of a network topology change. OSPF LSA (Link-State Advertisement) flooding occurs when there is an abrupt change in the network topology, causing the OSPF LSDB (Link-State Database) to become inconsistent across the network. The resulting BGP route churn and OSPF adjacency loss may be exacerbated by factors such as network congestion, misconfigured Cisco IOS settings, or hardware issues on the PE router or neighboring devices.

Pre-requisite Checks

- Pre-requisite: Verify the physical connectivity between HSR-PE-07 and neighboring devices.

- Pre-requisite: Check the redundant link status to ensure that the secondary link is operational.

- Pre-requisite: Dispatch remote hands to HSR Layout Access to perform any necessary site visits or device access.

Potential Root Causes

Cause	Probability
BGP Route Reflection Misconfiguration	High
OSPF LSA Flooding due to Network Topology Change	Medium
Cisco IOS Misconfiguration or Bug	Low
Hardware Issue on PE Router or Neighboring Devices	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP table version is 0, local router ID is 10.10.10.10, Description: This command displays a summary of the BGP table, including the number of neighbors, routes, and route refreshes. This helps to identify potential issues with BGP route reflection and route propagation.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays a list
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of OSPF neighbors, their state, and the interface used to reach them. This helps to identify potential issues with OSPF adjacency formation and LSA flooding.

Command: debug bgp 100, Expected Output: BGP debugging is on (level 5), Description: This command enables BGP debugging, which provides detailed information about BGP events, such as route refreshes and neighbor state changes. This helps to identify the root cause of the BGP route churn.
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Resolution Steps

Step 1: Step: Verify BGP Route Reflection Configuration, Command: show running-config | include router bgp, Description: This step involves verifying the BGP route reflection configuration on the PE router. The command shows the running configuration, and the include statement limits the output to the BGP configuration. Verify that the route reflector is configured correctly and that the client neighbors are correctly configured.

Step 2: Step: Clear OSPF LSDB, Command: clear ip ospf database, Description: This step involves clearing the OSPF LSDB on the PE router. This helps to reset the OSPF network and prevent further LSA flooding.

Step 3: Step: Reset BGP Neighbors, Command: clear ip bgp *, Description: This step involves resetting all BGP neighbors on the PE router. This helps to re-establish BGP connections and prevent further route churn.

Post-Check Verification

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Command: show ip bgp summary, Expected Output: BGP table version is 0, local router ID is 10.10.10.10, Description: This command displays a summary of the BGP table, including the number of neighbors, routes, and route refreshes. Verify that the BGP table is stable and that there are no further route changes.

Command: show ip ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays a list of OSPF neighbors, their state, and the interface used to reach them. Verify that the OSPF neighbors are stable and that there are no further adjacency changes.
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Roll-back Procedures

- Step: Restore Original BGP Configuration, Command: configure replace , Description: This step involves restoring the original BGP configuration on the PE router. This helps to ensure that the changes made during the troubleshooting process do not persist and cause further issues.

- Step: Revert OSPF LSDB, Command: clear ip ospf database, Description: This step involves reverting the OSPF LSDB on the PE router to its original state. This helps to prevent further LSA flooding and ensure network stability.

- Step: Re-establish BGP Connections, Command: clear ip bgp *, Description: This step involves re-establishing BGP connections on the PE router. This helps to prevent further route churn and ensure network stability.